

Modelling the Israel-Hamas War using Lanchester's Laws for Combat

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Foreword. This paper is an exploratory modelling exercise conducted jointly with Claude AI. The model structure and modelling decisions were directed by the author, while Claude AI assisted with mathematical derivation, parameter calibration, and analytical extension. The collaboration was genuinely iterative: Claude's outputs were repeatedly interrogated by the author, and several proposals — including dimensional assignments, effectiveness factor definitions, and validation claims — were found to be incorrect and had to be reworked through critical dialogue. The final model reflects the corrected outputs of this process, and illustrates both the utility and the limitations of large language models as analytical partners in operations research: capable of accelerating derivation and exploring model extensions, but prone to subtle mathematical errors and circular reasoning that require domain expertise to catch. Accordingly, the paper should be read as an experiment in AI-assisted military modelling rather than a validated predictive model.

Sections and tables marked with [†] denote content proposed by Claude AI without substantive iteration by the author. These include heuristic estimates and analytical extensions that were accepted on the basis of plausibility rather than independent verification.

1 Lanchester's Laws for Combat

In the early 20th century, Frederick W. Lanchester developed a mathematical methodology to leverage Ordinary Differential Equations (ODEs) to model force-on-force combat

as a function of time, with dependence on the combat strength and effectiveness of militaries. More interestingly (to me), is that Lanchester was also the co-inventor of the topic of operations research.

1.1 Lanchester's Linear Law

The **linear version** of Lanchester's Laws are relevant to linear tactics employed more historically. Imagine the 17th to 19th century warfare where soldiers fought in lines and columns, then the differential equations would be:

Let

- $R(t)$ denote the strength of the RED force at time t ,
- $B(t)$ denote the strength of the BLUE force at time t ,
- α denote the effectiveness of the BLUE force against RED,
- β denote the effectiveness of the RED force against BLUE.

Because of the assumption that linear warfare is a one-on-one combat, the Lanchester Linear Law is governed by the differential equations that reflect a constant casualty rate:

$$\frac{dR}{dt} = -\alpha, \quad (1.1)$$

$$\frac{dB}{dt} = -\beta. \quad (1.2)$$

Integrating gives the solutions

$$R(t) = R_0 - \alpha t, \quad (1.3)$$

$$B(t) = B_0 - \beta t, \quad (1.4)$$

where R_0 and B_0 are the initial strengths of the RED and BLUE forces, respectively. Eliminating time yields the combat invariant

$$\frac{R_0 - R}{\alpha} = \frac{B_0 - B}{\beta}, \quad (1.5)$$

or equivalently,

$$\beta(R_0 - R) = \alpha(B_0 - B). \quad (1.6)$$

For BLUE to prevail, the RED force must be completely eliminated while BLUE retains a positive fighting strength. Setting $R = 0$ in Equation (1.6) gives

$$\beta R_0 = \alpha(B_0 - B), \quad (1.7)$$

where $B > 0$ is the surviving BLUE force. Hence,

$$\alpha B_0 > \beta R_0, \quad (1.8)$$

or equivalently,

$$\frac{B_0}{R_0} > \frac{\beta}{\alpha}. \quad (1.9)$$

Under the Linear Law, the force ratio required for BLUE to prevail scales *linearly* with the relative combat effectiveness of the two forces. If $\alpha = \beta$, BLUE requires only a numerical advantage over RED.

1.2 Lanchester's Square Law

The **square version** of Lanchester's Laws is applicable to modern warfare, and was devised to consider the employment of ranged weapons such as rifles, artillery, missiles, and aircraft. Unlike linear combat, each soldier is no longer limited to engaging a single opponent. Instead, every combatant may simultaneously engage any target within range. Consequently, the casualty rate of each force is proportional to the size of the opposing force.

The Lanchester Square Law is therefore governed by the coupled differential equations

$$\frac{dR}{dt} = -\alpha B, \quad (1.10)$$

$$\frac{dB}{dt} = -\beta R. \quad (1.11)$$

Multiplying the first equation by βR and the second by αB gives

$$\beta R \frac{dR}{dt} = \alpha B \frac{dB}{dt}. \quad (1.12)$$

Recognising that

$$\frac{d(R^2)}{dt} = 2R \frac{dR}{dt}, \quad \frac{d(B^2)}{dt} = 2B \frac{dB}{dt},$$

we obtain

$$\frac{d}{dt} (\beta R^2 - \alpha B^2) = 0. \quad (1.13)$$

Integrating yields the combat invariant

$$\beta R^2 - \alpha B^2 = \beta R_0^2 - \alpha B_0^2, \quad (1.14)$$

or equivalently,

$$\beta (R^2 - R_0^2) = \alpha (B^2 - B_0^2). \quad (1.15)$$

For BLUE to prevail, the RED force must be completely eliminated while BLUE retains a positive fighting strength. Setting $R = 0$ in Equation (1.15) gives

$$\alpha B_0^2 > \beta R_0^2, \quad (1.16)$$

which may be rearranged as

$$\frac{B_0}{R_0} > \sqrt{\frac{\beta}{\alpha}}. \quad (1.17)$$

Unlike the Linear Law, the required force ratio depends on the square root of the relative combat effectiveness. Qualitatively, it reflects the multiplicative effects of combat power, where numerical strength is amplified by combat multipliers such as firepower, protection, mobility, and technology, giving larger forces a disproportionate advantage.

2 Israel-Hamas War

To model the Israel-Hamas (I-H) 2023 war mathematically, we begin by specifying the assumptions, structural considerations, and model parameters that define the system. These determine how combat is represented and what aspects of real conflict are abstracted away.

2.1 Assumptions

- We assume a closed, two-sided conflict between Israeli Defence Force (IDF), $I(t)$, and Hamas, $H(t)$, where each represents aggregated combat strength at time t .
- The model treats each force as homogeneous, meaning individual variation in training, morale, and equipment is subsumed into aggregated effectiveness parameters.
- Engagement is assumed to occur continuously over time, allowing the use of deterministic ordinary differential equations rather than discrete stochastic events.
- Hamas operates more like a guerrilla force, exploiting vast subterranean infrastructure and hiding within the civil dimension [2] to conduct surprise attacks on IDF. This can be represented in the model using Deitchman's extension of the Lanchester Model for Guerrilla Warfare [1].
- Hamas recruitment [6] is modelled as oscillatory — surging when organisational mass is sufficient to absorb and train new fighters, and collapsing when attrition degrades the cadre below the threshold needed to induct and deploy recruits, regardless of how many willing candidates exist in the population.

2.2 Parameters

Note that the symbols α and β are reused from Section 1 but with different definitions appropriate to the asymmetric Deitchman model.

Let α (day^{-1}) denote the IDF aimed-fire kill rate — the rate at which one IDF soldier kills one located Hamas fighter per day. Let γ (fighter^{-1}) denote the IDF search-and-detection rate — the fraction of Hamas fighters that one IDF soldier detects and renders engageable, capturing the effectiveness of ISR, intelligence networks, and targeting systems. Their product $\alpha\gamma$ ($\text{day}^{-1} \text{ fighter}^{-1}$) is the composite IDF kill rate per soldier per Hamas fighter per day.

Let β (day^{-1}) denote the Hamas area-fire kill rate — the rate at which one Hamas fighter kills IDF soldiers per day through ambush, IED, and indirect fire.

Since α and β share units of day^{-1} , their ratio α/β is dimensionless and represents the relative per-engagement lethality advantage of IDF aimed fire over Hamas area fire, independent of detection and force size.

2.3 ODEs

2.3.1 IDF attrition — Linear Law (Hamas area fire)

Hamas fighters do not aim at individual soldiers. Every guerrilla initiates area fire — rockets, IEDs, ambushes — at a rate that depends on Hamas organisational capacity, not on IDF strength. The IDF casualty rate therefore depends only on the number of guerrillas present, giving the Linear Law term:

$$\frac{dI}{dt} = -\beta H. \quad (2.1)$$

This is intentionally asymmetric: a rocket fired from a tunnel is equally lethal regardless of how many IDF soldiers are in the operational area. Only the number of Hamas fighters initiating fire determines the IDF casualty rate.

2.3.2 Hamas attrition — Square Law with oscillatory recruitment

IDF employs aimed fire in two sequential steps. First, IDF soldiers search and locate Hamas fighters at detection rate γ (fighter^{-1}), proportional to I — so the total detection rate across the force is γIH located fighters per unit time. **γ can also be interpreted as an intelligence and ISR multiplier**, since better ISR raises γ directly. Second, upon locating a fighter, IDF engages with kill rate α (day^{-1}). The composite kill rate is therefore $\alpha\gamma IH$ per day, giving the Square Law term. Hamas simultaneously recruits from the radicalised youth pool at an oscillatory rate $\rho(t)$, reflecting surges when organisational mass is sufficient to induct and deploy new fighters, and collapses when attrition degrades the cadre below the threshold needed to recruit effectively. This gives:

$$\frac{dH}{dt} = -\alpha\gamma I H + \rho(t) H, \quad (2.2)$$

where the recruitment rate is:

$$\rho(t) = r \left[1 + \sin\left(\frac{2\pi t}{T_r} + \phi\right) \right]. \quad (2.3)$$

The term $[1 + \sin(\cdot)]$ oscillates between 0 and 2, ensuring $\rho(t) \geq 0$ at all times. The product $\rho(t) \cdot H$ couples recruitment to surviving fighter strength: a heavily attrited Hamas cannot recruit effectively even when the oscillation is at its peak.

2.4 Solutions for IDF and Hamas Strength

With Claude AI's assistance in solving the ODEs, we obtain the following formal solutions in integral form. Since the integral in equation (2.5) depends on $H(\tau)$ through equation (2.4), and vice versa, the system has no closed-form solution and must be integrated numerically.

$$I(t) = I_0 - \beta \int_0^t H(\tau) d\tau \quad (2.4)$$

$$H(t) = H_0 \exp\left(-\alpha\gamma \int_0^t I(\tau) d\tau + r t + \frac{r T_r}{2\pi} \left[\cos \phi - \cos\left(\frac{2\pi t}{T_r} + \phi\right) \right]\right) \quad (2.5)$$

where r is the **base recruitment intensity** — the average daily rate¹ at which Hamas converts radicalised youth into fighters, expressed as a fraction of surviving fighter strength, when the recruitment cycle is at its mean level.

2.5 Combat Invariant and Victory Conditions

Multiplying equation (2.1) by $\alpha\gamma I$ and equation (2.2) by β :

$$\alpha\gamma I \frac{dI}{dt} = -\alpha\gamma\beta I H, \quad (2.6)$$

$$\beta \frac{dH}{dt} = -\alpha\gamma\beta I H + \beta\rho(t)H. \quad (2.7)$$

Subtracting equation (2.7) from equation (2.6) and recognising the left-hand side as an exact derivative:

$$\frac{d}{dt} \left(\frac{\alpha\gamma}{2} I^2 - \beta H \right) = -\beta\rho(t)H. \quad (2.8)$$

¹For example, if $r = 0.003$ per day and $H = 10,000$, Hamas inducts 30 new fighters per day on average before the oscillation modulates that up or down.

Since the right-hand side is non-zero, the quantity $\frac{\alpha\gamma}{2}I^2 - \beta H$ is not conserved — it is continuously driven by the recruitment term $\rho(t)H$. Integrating both sides from 0 to t and applying initial conditions $I(0) = I_0$, $H(0) = H_0$:

$$\frac{\alpha\gamma}{2}I^2 - \beta H = \frac{\alpha\gamma}{2}I_0^2 - \beta H_0 - \beta \int_0^t \rho(\tau) H(\tau) d\tau. \quad (2.9)$$

Let $I_f = I(t_f)$ and $H_f = H(t_f)$ denote the surviving strengths at the end of the engagement.

2.5.1 IDF victory

Setting $H_f = 0$ in equation (2.9) and requiring $I_f > 0$:

$$\frac{\alpha\gamma}{2}I_f^2 = \frac{\alpha\gamma}{2}I_0^2 - \beta H_0 - \beta \int_0^{t_f} \rho(\tau) H(\tau) d\tau > 0. \quad (2.10)$$

Rearranging:

$$I_0 > \sqrt{\frac{2\beta}{\alpha\gamma} \left(H_0 + \int_0^{t_f} \rho(\tau) H(\tau) d\tau \right)} \quad (2.11)$$

2.5.2 Hamas victory

Setting $I_f = 0$ in equation (2.9) and requiring $H_f > 0$:

$$-\beta H_f = \frac{\alpha\gamma}{2}I_0^2 - \beta H_0 - \beta \int_0^{t_f} \rho(\tau) H(\tau) d\tau < 0. \quad (2.12)$$

Rearranging:

$$H_0 + \int_0^{t_f} \rho(\tau) H(\tau) d\tau > \frac{\alpha\gamma}{2\beta} I_0^2 \quad (2.13)$$

The integral $\int_0^{t_f} \rho(\tau)H(\tau) d\tau$ represents the total fighters recruited over the campaign, accumulating faster during oscillation peaks of $\rho(t)$ and slower during troughs. The IDF victory condition requires I_0 to exceed the square root of Hamas's initial strength *plus* every fighter subsequently recruited — meaning operational tempo is decisive: **the longer the campaign runs, the larger the integral grows, and the harder IDF victory becomes.**

2.6 Combat Effectiveness Factor

[†] The CEF definition and its restatement of the victory condition were proposed by Claude AI.

The solutions (2.4) and (2.5) embed two attrition processes: $\beta \int_0^{t_f} H d\tau$ drives total IDF casualties, and $\alpha\gamma \int_0^{t_f} I d\tau$ drives total Hamas casualties. A natural, dimensionless measure of relative effectiveness is the ratio of the per-fighter kill rates as they appear in the two solutions, evaluated at the mean force levels $\bar{I}$ and $\bar{H}$:

$$\text{CEF} = \frac{\alpha\gamma \bar{I}}{\beta}. \quad (2.14)$$

Checking dimensions:

$$\left[\frac{\alpha\gamma \bar{I}}{\beta} \right] = \frac{\text{day}^{-1} \text{fighter}^{-1} \times \text{fighters}}{\text{day}^{-1}} = \text{dimensionless}. \checkmark \quad (2.15)$$

CEF is the ratio of the rate at which each Hamas fighter is killed (numerator: $\alpha\gamma \bar{I}$, the aimed-fire attrition rate evaluated at mean IDF strength) to the rate at which each Hamas fighter kills IDF soldiers (denominator: β). When $\text{CEF} > 1$, IDF kills Hamas fighters faster per Hamas fighter than Hamas kills IDF soldiers per Hamas fighter; equivalently, CEF is the **exchange ratio** — Hamas casualties per IDF casualty. The IDF victory condition (2.11) can be rewritten in terms of CEF by substituting $\alpha\gamma = \text{CEF} \cdot \beta / \bar{I}$:

$$I_0 > \sqrt{\frac{2\bar{I}}{\text{CEF}}} H_{\text{eff}}, \quad (2.16)$$

where $H_{\text{eff}} = H_0 + \int_0^{t_f} \rho H d\tau$ is the effective Hamas strength including all recruitment.

Unlike the Square Law where $\text{CEF} = \alpha/\beta$ compares per-fighter kill rates directly, the Deitchman $\text{CEF} = \alpha\gamma \bar{I}/\beta$ incorporates IDF force size $\bar{I}$ and the detection rate γ . This reflects the structural asymmetry of the model: IDF effectiveness scales with I through the search process, so the same α at a larger $\bar{I}$ produces a proportionally higher effective kill rate. Firepower and mass are not substitutable at a fixed rate — they multiply.

2.7 Parameter Calibration

Parameters are calibrated over the ground phase of the conflict from 27 October 2023 to the ceasefire of 19 January 2025, a period of approximately $T = 450$ days. The empirical anchors are given in Table 2.1.

Table 2.1: Empirical anchors for parameter calibration

Quantity	Value	Source
IDF peak ground commitment I_0	70,000	[5]
Hamas initial strength H_0	25,000	[6]
IDF killed in action	≈ 800	[7]
Hamas fighters killed	$\approx 15,000^2$	[3, 4]
Hamas fighters recruited	$\approx 12,500$	[6]
Ground operation duration T	450 days	[8]

2.7.1 Step 1 — Calibrating β from IDF casualties

From equation (2.4), cumulative IDF casualties over T days satisfy

$$\text{IDF KIA} \approx \beta \cdot \bar{H} \cdot T, \quad (2.17)$$

where $\bar{H}$ is the mean Hamas strength. Hamas's final strength is estimated as $H_f \approx 25,000 - 15,000 + 12,500 = 22,500$, giving $\bar{H} \approx (25,000 + 22,500)/2 = 23,750$. Substituting:

$$\beta = \frac{800}{23,750 \times 450} \approx 7.5 \times 10^{-5} \text{ day}^{-1}. \quad (2.18)$$

2.7.2 Step 2 — Calibrating $\alpha\gamma$ from Hamas casualties

From equation (2.5), cumulative Hamas attrition satisfies

$$\text{Hamas KIA} \approx \alpha\gamma \cdot \bar{I} \cdot \bar{H} \cdot T, \quad (2.19)$$

where $\bar{I} \approx (70,000 + 69,200)/2 = 69,600$. Solving:

$$\alpha\gamma = \frac{15,000}{69,600 \times 23,750 \times 450} \approx 2.0 \times 10^{-8} \text{ day}^{-1} \text{ fighter}^{-1}. \quad (2.20)$$

2.7.3 Step 3 — Computing CEF and decomposing γ

Substituting the calibrated values into equation (2.14):

$$\text{CEF} = \frac{\alpha\gamma \bar{I}}{\beta} = \frac{2.0 \times 10^{-8} \times 69,600}{7.5 \times 10^{-5}} \approx 18.6. \quad (2.21)$$

This matches the empirical kill ratio of $15,000/800 \approx 18.75 : 1$ by construction, since $\alpha\gamma$ and β were both calibrated from the same casualty data. The agreement reflects internal consistency rather than independent validation; predictive validity would require testing against a held-out period or a separate conflict.

[†]The capability factor decomposition in Table 2.2 was proposed by Claude AI as a heuristic estimate. The individual factor values are informed by open-source capability assessments but are not independently calibrated from empirical engagement data.

To decompose γ separately, the capability factors in Table 2.2 give the dimensionless lethality ratio α/β :

$$\frac{\alpha}{\beta} = 4.5 \times 3.0 \times 0.6 \times 2.0 \times 0.4 = 6.48, \quad (2.22)$$

²Taken as the mean between the IDF's classified internal database, which listed 8,900 named fighters as confirmed or probably dead as of May 2025 per a joint investigation by The Guardian, +972 Magazine, and Local Call [3], and the IDF's peak public claim of approximately 20,000 fighters killed [4]. The mean of these two figures is $\approx 14,450$, rounded to 15,000.

so $\alpha = 6.48 \times \beta = 6.48 \times 7.5 \times 10^{-5} \approx 4.9 \times 10^{-4} \text{ day}^{-1}$, and:

$$\gamma = \frac{\alpha\gamma}{\alpha} = \frac{2.0 \times 10^{-8}}{4.9 \times 10^{-4}} \approx 4.1 \times 10^{-5} \text{ fighter}^{-1}. \quad (2.23)$$

A detection rate of $\gamma \approx 4.1 \times 10^{-5} \text{ fighter}^{-1}$ means each IDF soldier detects and enables engagement of only 4.1×10^{-5} Hamas fighters per unit of Hamas strength — a very low value reflecting the severe constraint imposed by tunnel concealment, civilian embedding, and dense urban terrain on IDF targeting effectiveness.

Table 2.2: Capability factor decomposition: ratio of kill rates α/β (dimensionless)[†]

Factor	Value	Rationale
Firepower	4.5	Precision airstrikes, tank guns, drones vs RPGs and small arms
Survivability	3.0	Trophy APS, Merkava armour vs Hamas anti-armour
Mobility	0.6	Urban terrain heavily penalises IDF vehicle advantage
ISR/C2	2.0	IDF full ISR suite vs Hamas distributed cell structure
Terrain	0.4	Dense urban + tunnel network — most severe terrain penalty
α/β	6.48	Product of all factors (dimensionless)

2.7.4 Step 4 — Calibrating r from recruitment data

Since the mean value of the oscillation $[1 + \sin(\cdot)]$ is unity, cumulative recruitment satisfies

$$\text{Hamas recruited} \approx r \cdot \bar{H} \cdot T, \quad (2.24)$$

giving

$$r = \frac{12,500}{23,750 \times 450} \approx 1.17 \times 10^{-3} \text{ day}^{-1}. \quad (2.25)$$

2.7.5 Calibrated parameters

Table 2.3: Calibrated model parameters

Parameter	Value	Units	Interpretation
β	7.5×10^{-5}	day^{-1}	Hamas area-fire kill rate per fighter per day
α	4.9×10^{-4}	day^{-1}	IDF aimed-fire kill rate per soldier per located target per day
γ	4.1×10^{-5}	fighter^{-1}	IDF detection rate per soldier per unit Hamas strength
$\alpha\gamma$	2.0×10^{-8}	$\text{day}^{-1} \text{ fighter}^{-1}$	Composite IDF kill rate
CEF	18.6	dimensionless	Solution-derived exchange ratio = $\alpha\gamma\bar{I}/\beta$
r	1.17×10^{-3}	day^{-1}	Base Hamas recruitment intensity

2.8 Relative Combat Power Analysis

Approximating the recruitment integral by the empirical total of 12,500 fighters gives $H_{\text{eff}} = H_0 + 12,500 = 37,500$. From equation (2.16):

$$I_{\min} = \sqrt{\frac{2\bar{I}}{\text{CEF}}} H_{\text{eff}} = \sqrt{\frac{2 \times 69,600}{18.6}} \times 37,500 \approx 16,760 \text{ fighters.} \quad (2.26)$$

With $I_0 = 70,000$, the IDF force ratio is

$$\frac{I_0}{I_{\min}} \approx 4.17. \quad (2.27)$$

The IDF committed approximately **4.2 times** the minimum force required for victory. Without Hamas recruitment the threshold falls to 13,700, a ratio of 5.1. Recruitment reduced the effective IDF force margin by **22.5%**, consistent with Blinken’s assessment that Hamas recruited “almost as many fighters as it lost” [6].

2.9 Prediction of Next Major Conflict

[†] *The secular trend framework and resurgence prediction below were proposed by Claude AI as an analytical extension of the calibrated model. The post-ceasefire force level I_m , the trough estimate H_{trough} , and the re-engagement threshold H^* are assumptions, not observed quantities.*

2.9.1 Secular trend

Following the ceasefire, the IDF withdrew to the “Yellow Line” demarcating Israeli-controlled territory, maintaining an estimated force of $I_m \approx 30,000$ across several divisions [9]. The term “secular” is borrowed from economics and demography, where it describes a long-run directional tendency stripped of short-term fluctuations. Here, Hamas’s strength oscillates daily due to $\rho(t)$, but the question of whether Hamas is growing or shrinking *on average* is determined by the trend beneath those oscillations. Define the secular trend parameter

$$\Delta = r - \alpha\gamma I_m, \quad (2.28)$$

which captures the net daily growth rate of Hamas at mean recruitment. This follows directly from equation (2.2): at mean recruitment $\rho = r$ and constant I_m , setting $dH/dt = 0$ gives the breakeven condition $r = \alpha\gamma I_m$, of which Δ measures the signed departure. **When $\Delta > 0$, recruitment outpaces attrition and Hamas grows; when $\Delta < 0$, Hamas declines.** Substituting:

$$\Delta = 1.17 \times 10^{-3} - (2.0 \times 10^{-8})(30,000) = +5.7 \times 10^{-4} \text{ day}^{-1}. \quad (2.29)$$

Since $\Delta > 0$, Hamas grows under monitoring conditions, consistent with the US intelligence assessment that Hamas “regroups and re-emerges each time Israel pulls back” [6].

2.9.2 Time to resurgence

With Hamas at an estimated $H_{\text{trough}} \approx 5,000$ after the ceasefire³ and a re-engagement threshold of $H^* \approx 15,000$ — approximately 60% of pre-war strength — the time to reconstitute under net growth rate Δ is

$$t^* = \frac{1}{\Delta} \ln \frac{H^*}{H_{\text{trough}}} = \frac{\ln 3}{5.7 \times 10^{-4}} \approx 1,928 \text{ days.} \quad (2.30)$$

The model predicts Hamas will reconstitute to the re-engagement threshold approximately **1,928 days** ($\approx$ **5.3 years**) after the January 2025 ceasefire, placing the next major Israeli ground operation around **mid-2030**. This is broadly consistent with the historical inter-war interval of 2–5 years across the 2008, 2012, 2014, 2021, and 2023 conflicts, though at the upper end, reflecting the larger sustained IDF presence of $I_m = 30,000$ compared to previous inter-war periods.

2.9.3 Critical monitoring force

Setting $\Delta = 0$ in equation (2.30):

$$I^* = \frac{r}{\alpha\gamma} = \frac{1.17 \times 10^{-3}}{2.0 \times 10^{-8}} \approx 58,500 \text{ fighters.} \quad (2.31)$$

Permanent suppression requires sustaining approximately **58,500 IDF troops** in Gaza indefinitely — approaching the peak ground commitment of 70,000, and formalising mathematically why a purely military solution is strategically unsustainable.

3 Model Limitations

The following limitations should be considered when interpreting the results.

3.1 Structural simplifications

The model treats each force as homogeneous, aggregating infantry, armour, special operations, and air support into a single scalar $I(t)$. In reality, IDF combat power is a function of *which* units are deployed — an armoured brigade contributes differently from a reserve infantry brigade — and unit rotations (documented throughout the campaign [9]) change the qualitative character of the force without changing its numerical strength. Hamas is similarly aggregated, collapsing the distinction between trained Qassam Brigade fighters, recently recruited operatives, and affiliated militia into a single $H(t)$.

³Estimated from the calibrated $H_f \approx 22,500$, discounted by the assumption that the ceasefire further degraded Hamas's deployable strength through continued Israeli targeted operations and organisational attrition during the transition period.

3.2 Deterministic framework

The model uses deterministic ODEs, meaning it produces a single trajectory rather than a distribution of outcomes. Real combat is stochastic: a single event (a leader killed, a tunnel discovered, a friendly-fire incident) can shift the trajectory discontinuously. The model has no mechanism for such shocks. A Monte Carlo extension wrapping the ODE system in randomised parameter draws would partially address this, but was not implemented in the present analysis.

3.3 Constant parameters

The attrition coefficients α , β , and γ are calibrated as campaign averages and held constant. In practice, γ almost certainly declined as Hamas shifted from conventional defence to distributed cell warfare in mid-2024, and β may have increased as Hamas transitioned to IED-heavy tactics. A time-varying parameter model would better capture these dynamics but would require sub-period casualty data that is not reliably available.

3.4 Closed-system assumption

The model assumes a closed two-sided conflict. It excludes third-party interventions (Hezbollah, Houthi, and Iranian actions that diverted IDF resources), international political constraints on IDF operations (ceasefire pressures, aid corridor requirements), and the humanitarian dimension that shaped both the operational environment and the political will to sustain operations.

3.5 Victory defined as annihilation

The victory conditions require one side to reach zero, which is not how real conflicts end. The January 2025 ceasefire occurred with both forces intact — Hamas degraded but surviving, IDF withdrawing for political rather than military reasons. The model’s “victory” is a mathematical abstraction; the actual conflict outcome was determined by political thresholds that the model does not represent.

3.6 Recruitment function

The oscillatory recruitment function $\rho(t)$ imposes a fixed period T_r and phase ϕ that are not calibrated from data — they are structural assumptions. The actual recruitment pattern is likely driven by discrete events (ceasefires enabling reorganisation, Israeli withdrawals from specific areas, surges in radicalisation following high-profile attacks) rather than a smooth sinusoidal oscillation. The function captures the qualitative pattern of surging and collapsing recruitment but not its event-driven timing.

3.7 Circular calibration

As noted in Section 2.7, the CEF of 18.6 reproduces the empirical kill ratio by construction, not by independent prediction. The model’s predictive power is limited to its

out-of-sample extensions — principally the resurgence timeline and critical monitoring force — which depend on the calibrated parameters holding in a post-ceasefire regime that may differ structurally from the wartime environment in which they were estimated.

3.8 Capability table

The factor decomposition in Table 2.2 is a heuristic estimate proposed by Claude AI. The individual values (Firepower = 4.5, Survivability = 3.0, etc.) are informed by qualitative capability assessments but are not derived from engagement-level data. Different plausible factor choices would yield different α/β ratios and hence different γ values, while leaving the composite $\alpha\gamma$ and all downstream results unchanged — since $\alpha\gamma$ is calibrated directly from casualty data independently of the table.

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